

PRACTICAL -3:

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nirvighna@DESKTOP-VS57D2  ×  +  ▾  -  □  ×
GNU nano 7.2                  process_states.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Starting Process State Demonstration\n");
    printf("Original Process PID: %d\n\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    /* Child process */
    if (pid == 0)
    {
        printf("----- CHILD PROCESS -----\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());
        printf("Child is now running.\n");

        sleep(15);

        printf("Child process is terminating.\n");
        exit(0);
    }

    /* Parent process */
    else

```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$ nano process_states.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$ ls -l process_states.c
-rw-r--r-- 1 nirvighna nirvighna 1058 Sep 25 07:41 process_states.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1/src$ |
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin/process_states
-rwxr-xr-x 1 nirvighna nirvighna 18056 Sep 25 07:42 bin/process_states
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/process_states
Starting Process State Demonstration
Original Process PID: 1496
```

```
----- PARENT PROCESS -----
Parent PID : 1496
Child PID : 1497
Parent is waiting for child...
----- CHILD PROCESS -----
Child PID : 1497
Child PPID : 1496
Child is now running.
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ps -ef | grep process_states
nirvigh+ 1504 315 0 07:44 pts/0 00:00:00 grep --color=auto proces
s_states
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ps -o pid,ppid,state,stat,cmd -C proces
s_states
PID PPID S STAT CMD
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ps -o pid,ppid,state,stat,cmd -C proces
s_states,ppid,state,stat,cmd -C process_states
PID PPID S STAT CMD
2099 315 S S ./bin/process_states
2100 2099 S S ./bin/process_states
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ Child process is terminating.
Child has terminated.
Parent process is terminating.
Child PID : 5235
```